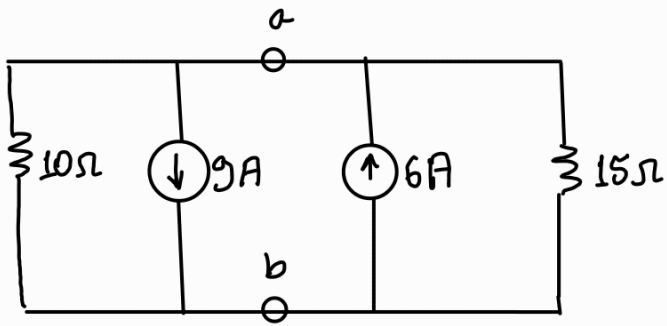
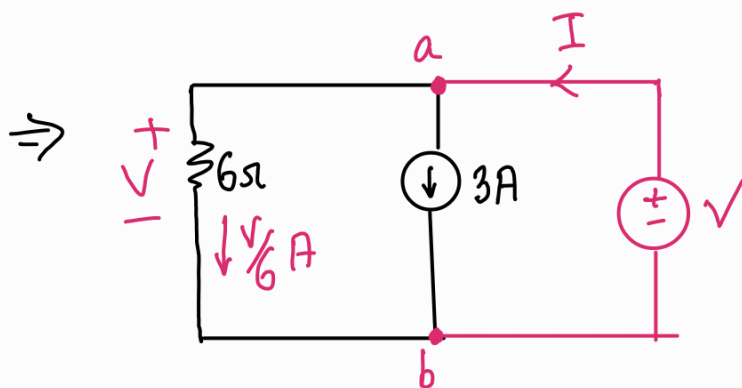
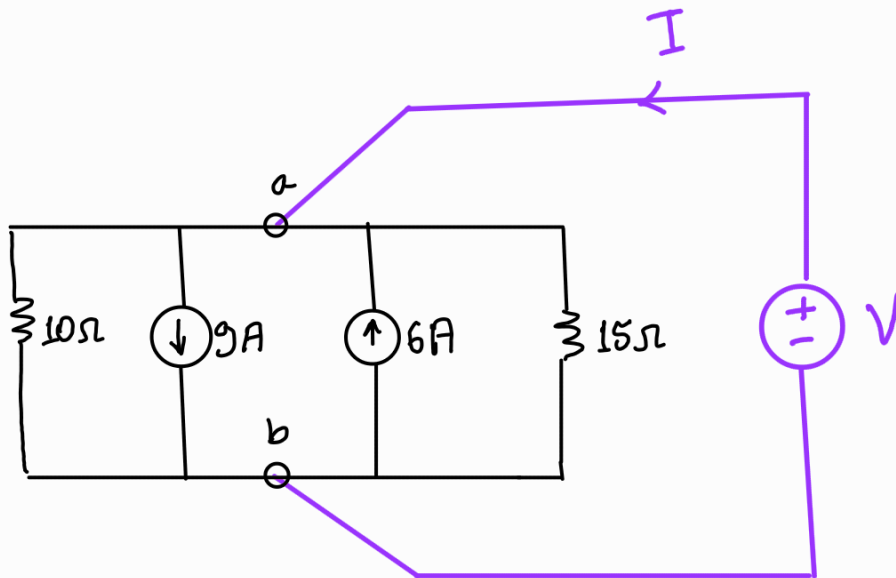


Problem 1



Derive I-V characteristics of the following circuit with respect to the terminals a-b

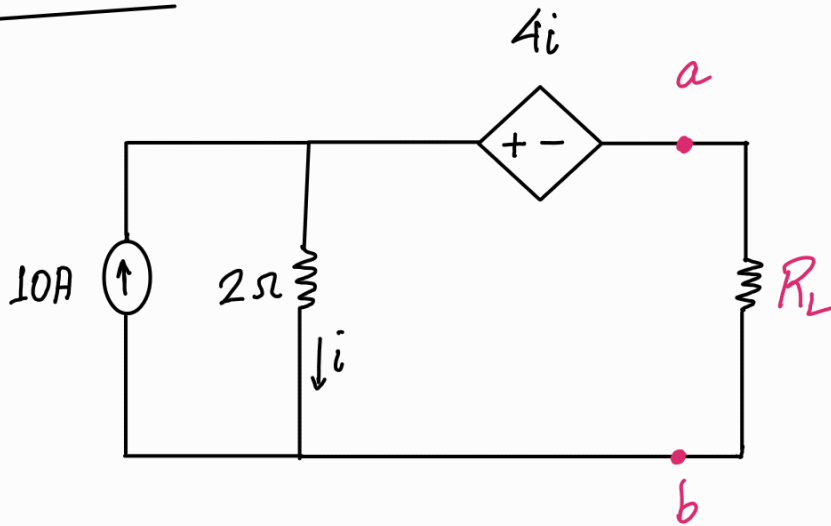


KCL @ a —

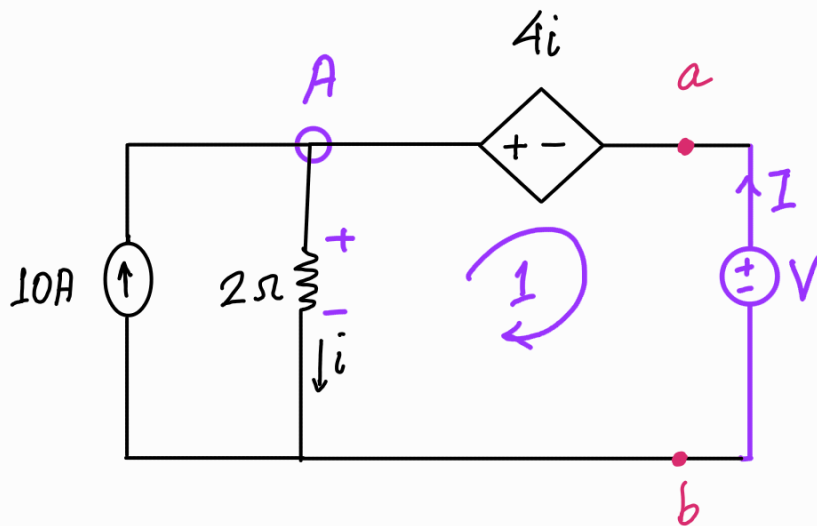
$$I = \frac{V}{6} + 3$$

Solution

Problem 2



Derive the I-V characteristics of the portion left to the terminals a-b.



$$\text{KCL @ A} - 10 + I = i \text{ ———— (i)}$$

$$\text{KVL @ 1} - V - 2i + 4i = 0$$

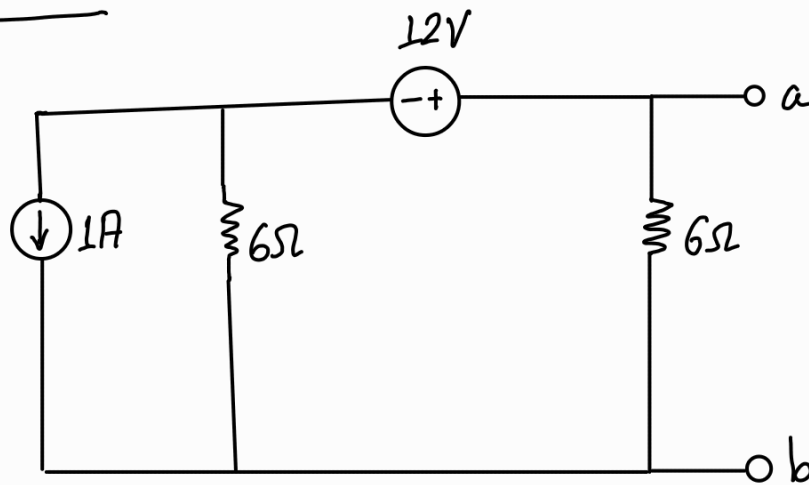
$$\Rightarrow V = -2i$$

$$\Rightarrow i = -\frac{V}{2} \text{ ———— (ii)}$$

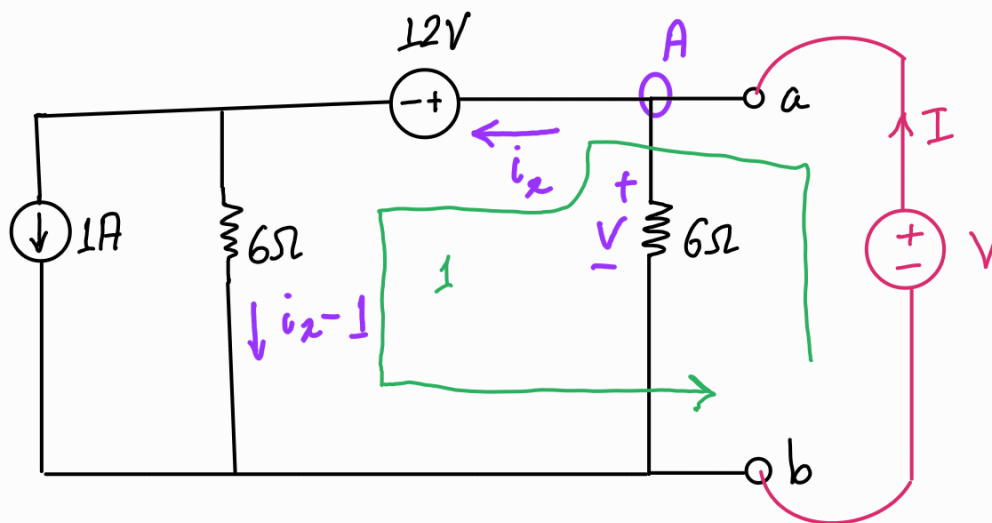
$$\therefore \text{(i)} \rightarrow \boxed{I = -\frac{V}{2} - 10}$$

↪ Solution

Problem 3



From the following circuit derive the I-V characteristics between the terminals a-b



$$\text{KCL @ A} \rightarrow I = i_x + \frac{V}{6} \quad \text{--- (i)}$$

$$\text{KVL @ 1} \rightarrow 12 + 6i_x - 6 - V = 0$$

$$\Rightarrow 6i_x = V - 6$$

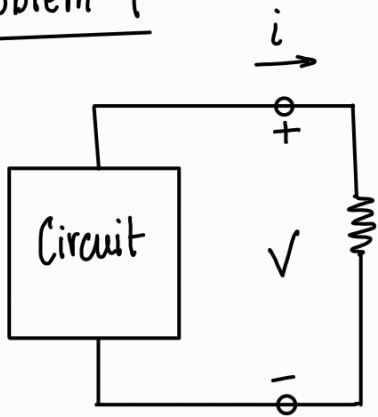
$$\Rightarrow i_x = \frac{V}{6} - 1$$

$$\therefore \text{(i)} \rightarrow I = \frac{V}{6} - 1 + \frac{V}{6}$$

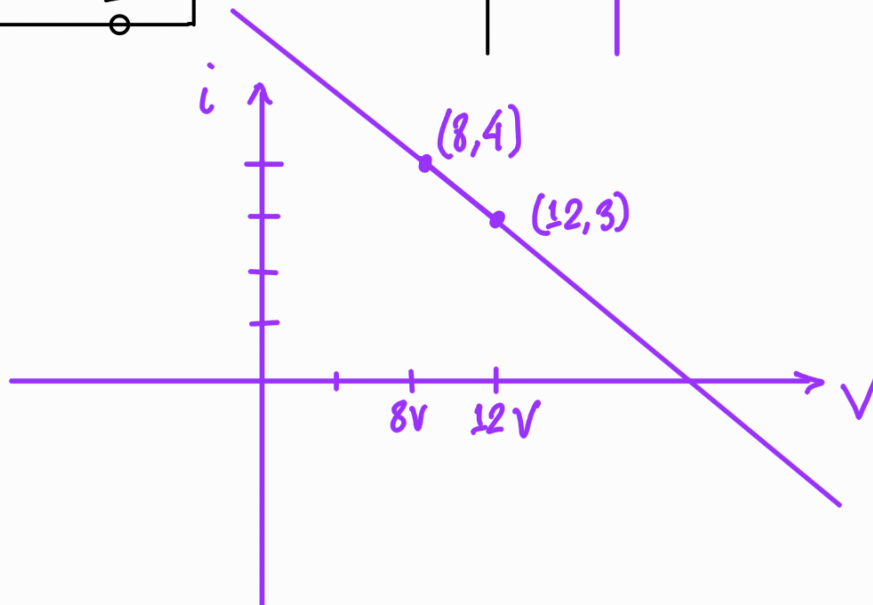
$$\Rightarrow \boxed{I = \frac{V}{3} - 1}$$

→ Solution

Problem 4



R	i	$V=iR$
$2k\Omega$	$4mA$	$8V$
$4k\Omega$	$3mA$	$12V$



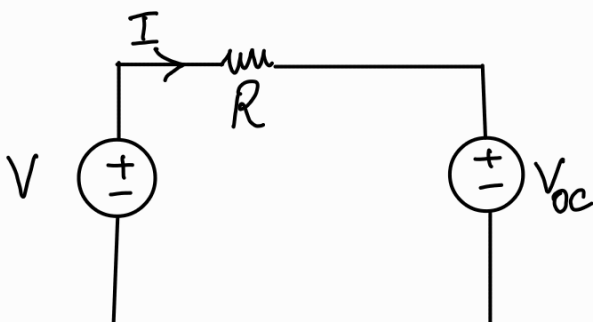
Equation from graph — $\frac{4-3}{8-12} = \frac{I-3}{V-12}$

$$\Rightarrow V-12 = -4I+12$$

$$\Rightarrow I = -\frac{V}{4} + 6$$

Now how do we find the simplest form of ckt from I-V equations?

The simplest form of ckt —



Simplest IV equation

$$\rightarrow I = \frac{V-V_{oc}}{R}$$

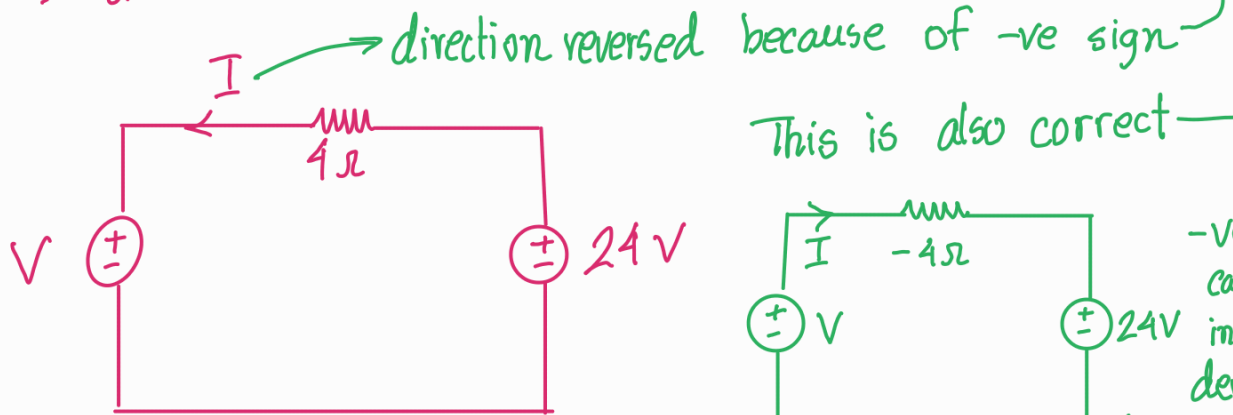
To draw the ckts we need to bring the I-V equation to this form.

Going back to our problem —

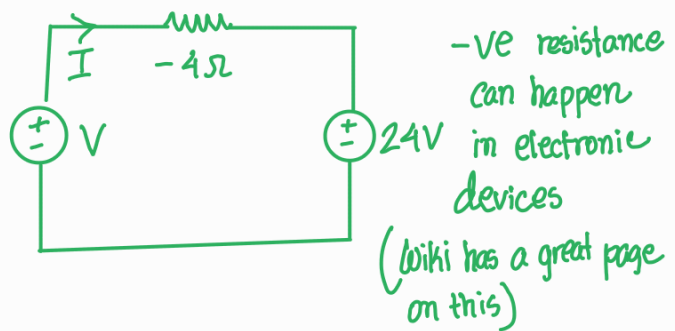
our IV equation $\rightarrow I = -\frac{V}{4} + 6$

$$= -\frac{V-24}{4}$$

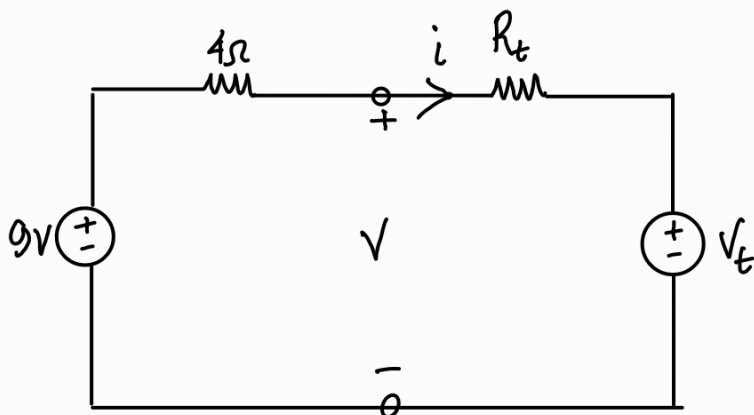
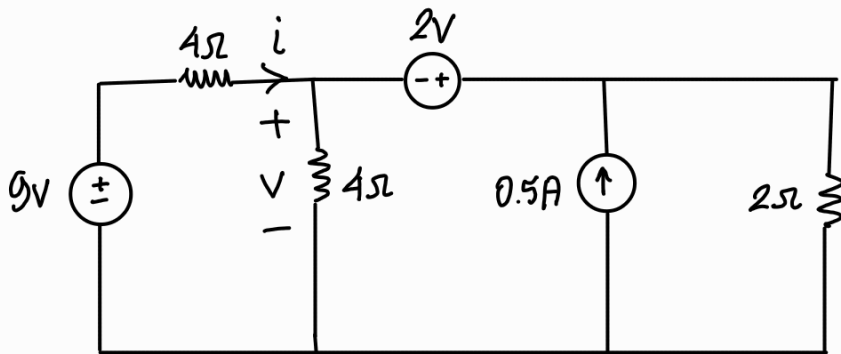
$\therefore$ ckt —



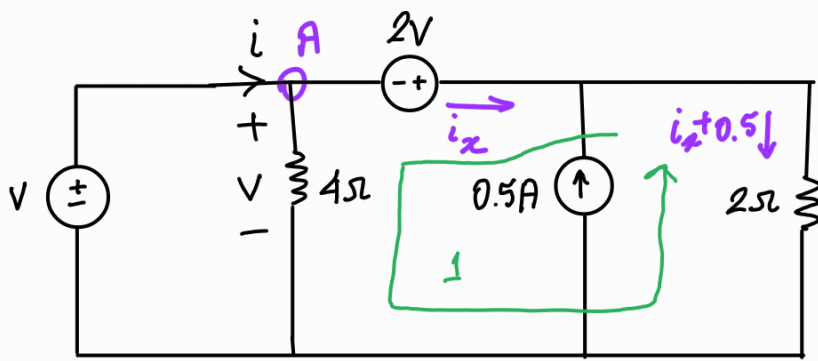
This is also correct —



Problem 5



Determine the value of R_t & V_t if the two circuits are equivalent to each other.



We have to at first find the I-V characteristics equation for this circuit.

$$\text{KCL @ A} \rightarrow i = \frac{V}{4} + i_x \quad \text{--- (i)}$$

$$\text{KVL @ 1} \rightarrow V - 2i_x - 1 + 2 = 0$$

$$\Rightarrow i_x = \frac{V}{2} + \frac{1}{2}$$

$$\therefore \text{(i)} \rightarrow i = \frac{V}{4} + \frac{V}{2} + \frac{1}{2}$$

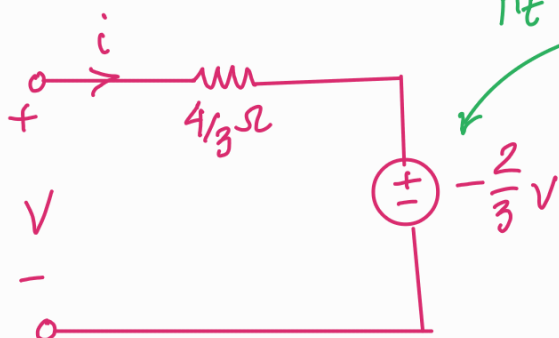
$$= \frac{3V}{4} + \frac{1}{2}$$

now we have to change it to the simplest form—

$$i = \frac{3V+2}{4}$$

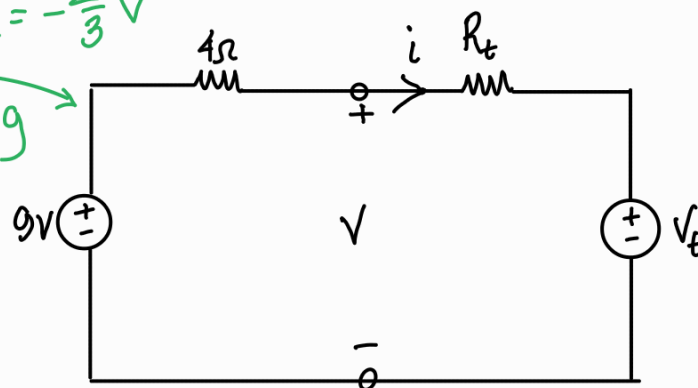
$$= \frac{V - (-\frac{2}{3})}{4/3}$$

$\therefore$ the simplest ckt —

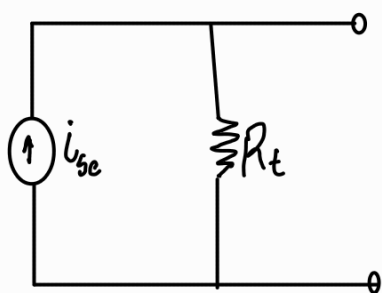
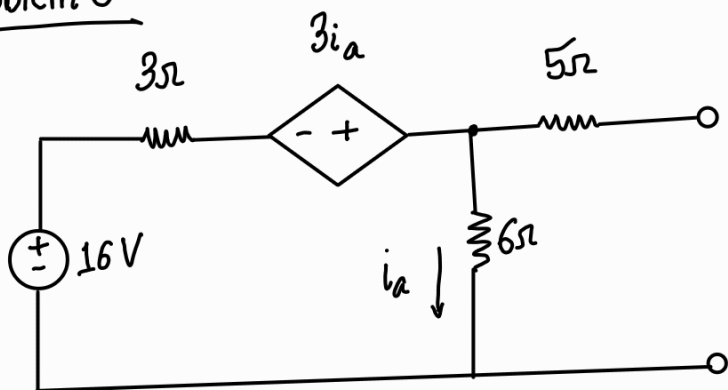


$$R_t = \frac{4}{3} \Omega ; V_t = -\frac{2}{3} V$$

Comparing

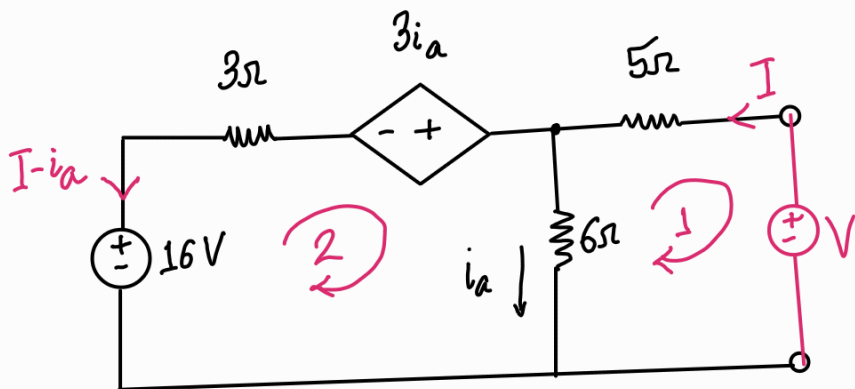


Problem 6



Determine the value of i_{sc} and R_t if two ckt's are equivalent.

Find the I-V characteristics equation for the original ckt —



KVL @ 1 —

$$V - 6i_a - 5I = 0 \quad \text{--- (1)}$$

KVL @ 2 —

$$-16 - 3I + 3i_a - 3i_a + 6i_a = 0$$

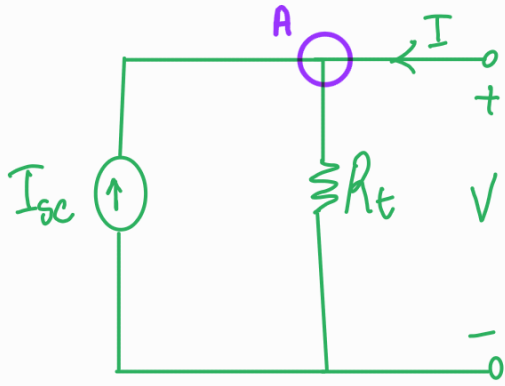
$$\Rightarrow 6i_a = 16 + 3I$$

$$\text{--- (1)} \Rightarrow V - 16 - 3I - 5I = 0$$

$$\Rightarrow I = \frac{V}{8} - 2$$

We cannot use our previous simplest circuit because it was in the form of voltage source.

Lets derive a simplest equation form for the ckt —



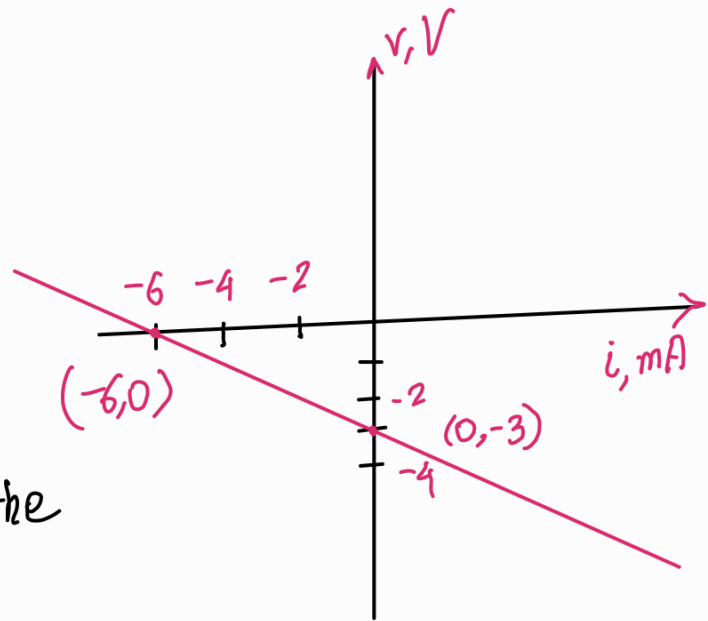
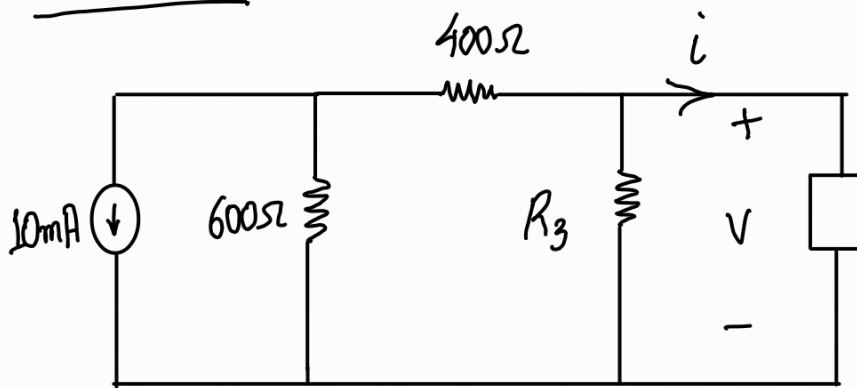
KCL @ A —

$$I + I_{sc} = \frac{V}{R_t}$$

$$\Rightarrow I = \frac{V}{R_t} - I_{sc}$$

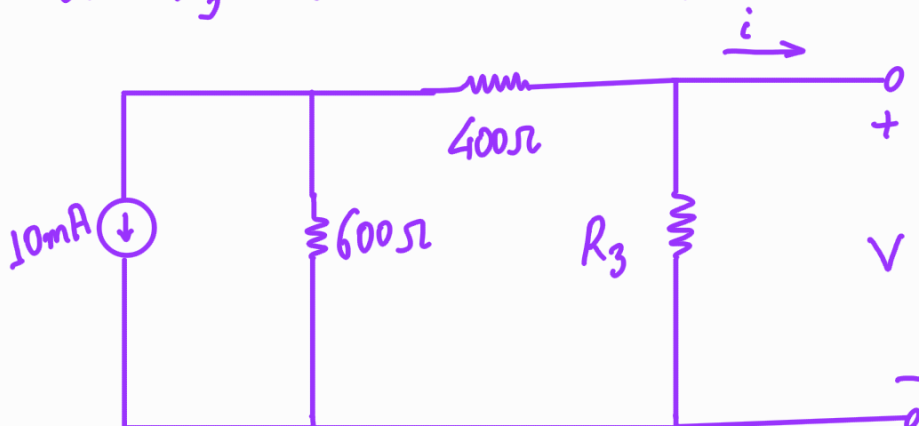
Now our original I - V equation — $I = \frac{V}{8} - 2$ comparing $I_{sc} = 2A$
 $R_t = 8\Omega$

Problem 7



From the v - i characteristics find the value of R_3

Since R_3 is on the left side, we are only concerned with that part

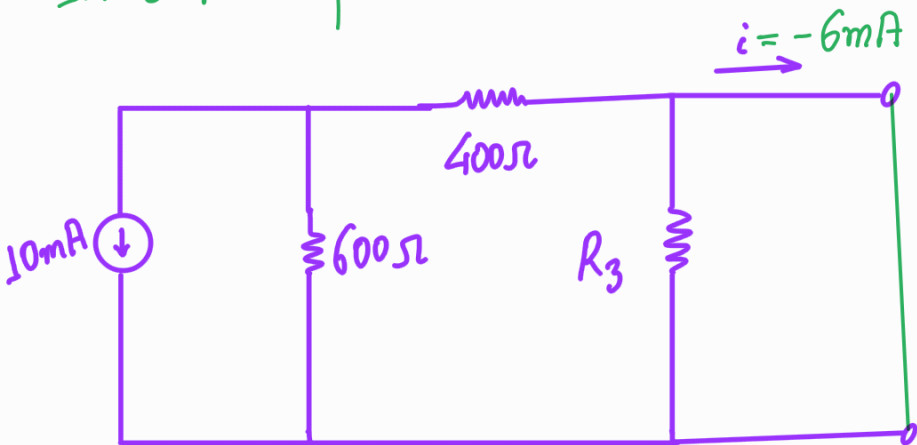


In such problems, we will take advantage of the extreme points in the graph.

1st extreme point $\rightarrow (-6\text{mA}, 0\text{V}) \rightarrow$ short circuit

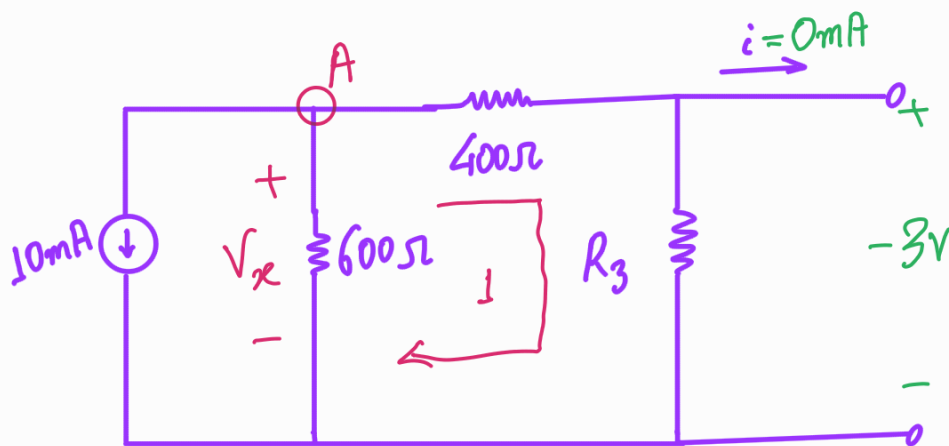
2nd extreme point $\rightarrow (0\text{mA}, -3\text{V}) \rightarrow$ open circuit

1st extreme point ckt —



Cannot be used because we lose R_3 to a short circuit

2nd extreme point ckt —



We can use this ckt.

$$\text{KCL @ A} \rightarrow -10 + \frac{V_x}{600} + \frac{-3}{R_3} = 0 \quad \text{--- (i)}$$

$$\text{Voltage divider in 1} \rightarrow -3 = \frac{R_3 V_x}{400 + R_3}$$

$$\Rightarrow -1200 - 3R_3 = R_3 V_x$$

$$\Rightarrow V_x = -\frac{1200}{R_3} - 3$$

$$\therefore \textcircled{i} \Rightarrow 10^{-2} - \frac{2}{R_3} - \frac{1}{200} - \frac{3}{R_3} = 0$$

$$\Rightarrow \frac{5}{R_3} = -\frac{2-1}{200}$$

$$\Rightarrow R_3 = 1000 = 1\text{K}\Omega$$